UNIVERSIDAD DE HUELVA

MÁSTER EN INGENIERÍA INFORMÁTICA

Universidad de Huelva
Escuela Técnica Superior de Ingeniería

Tarea CaaS Contenerizar Aplicación

Ciclo lectivo 2023-24

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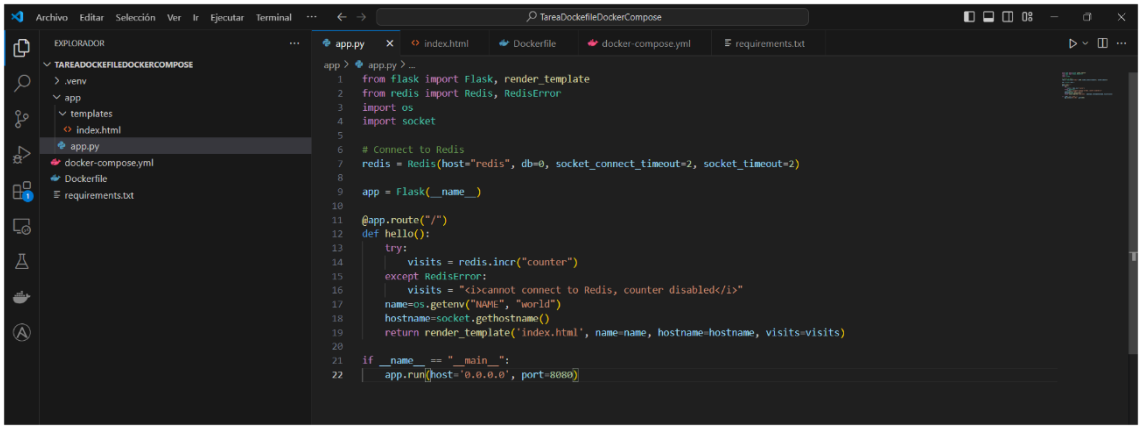
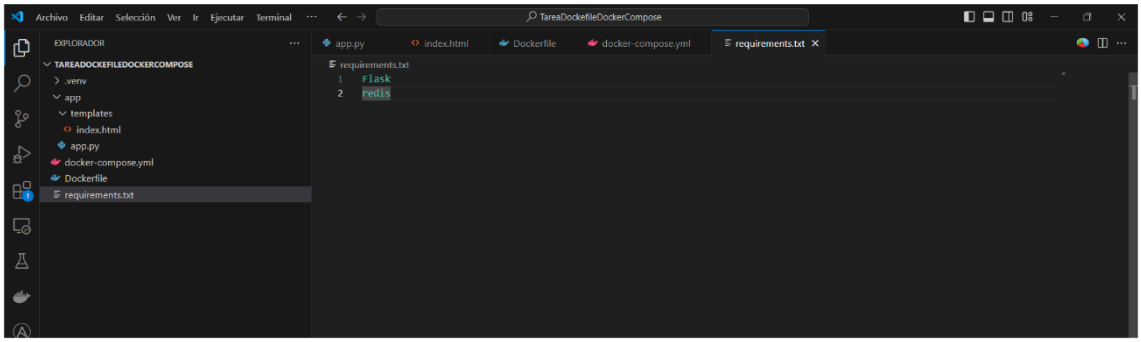
Fecha de Entrega: 06/06/2024

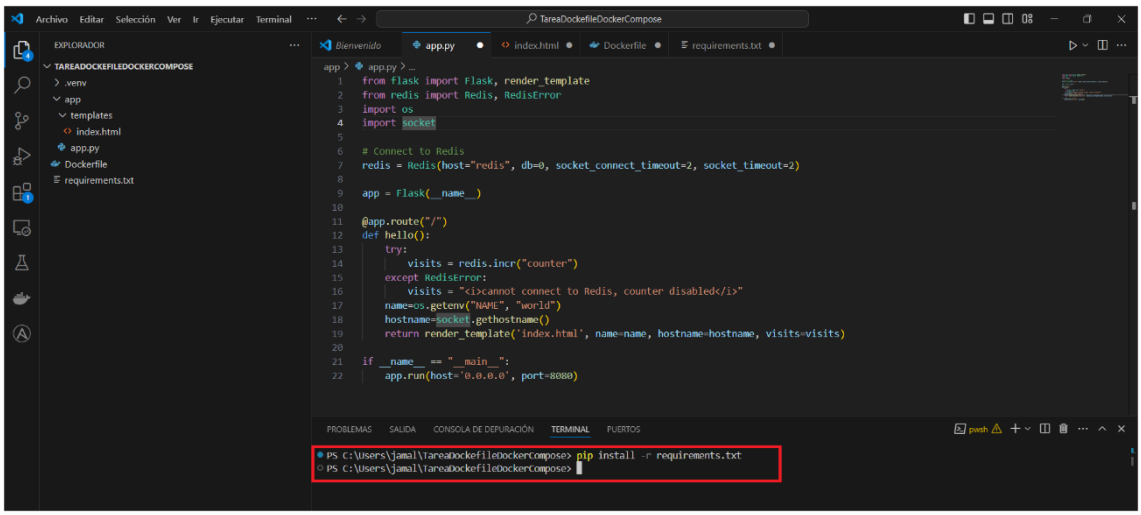
crear fichero app.py



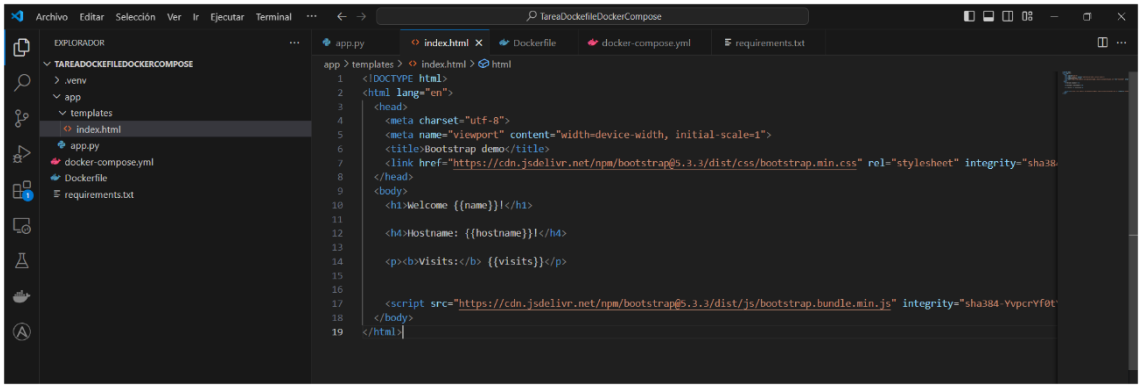
crear un fichero requirements.txt



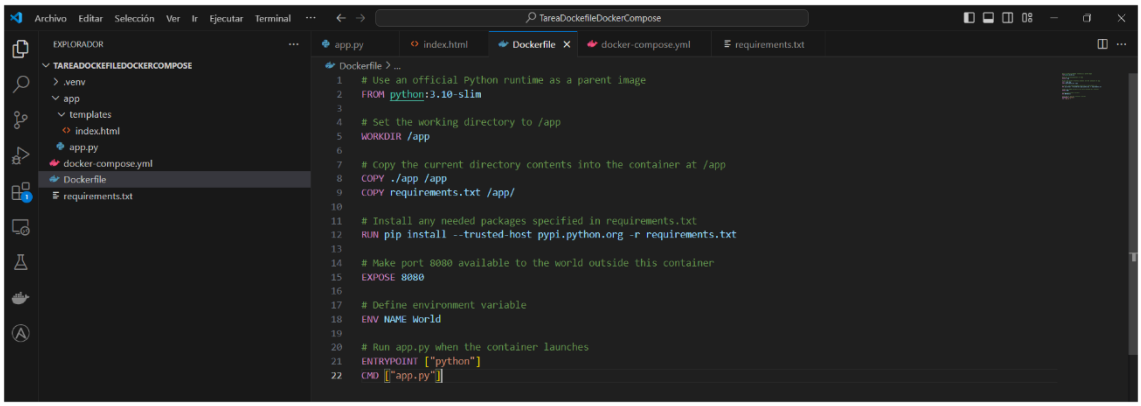
En un entorno de Python se instalarían con



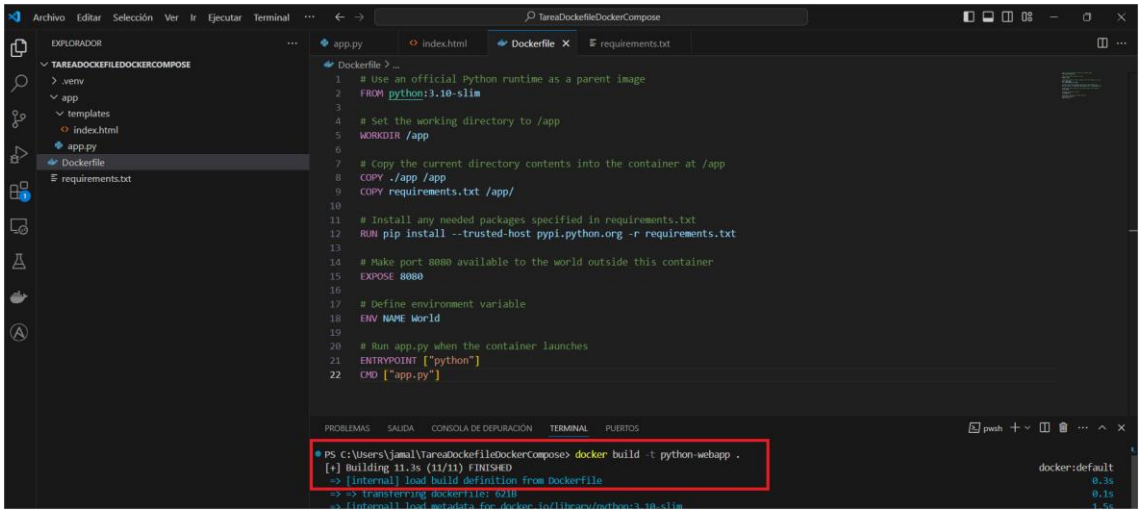
crear la vista index.html



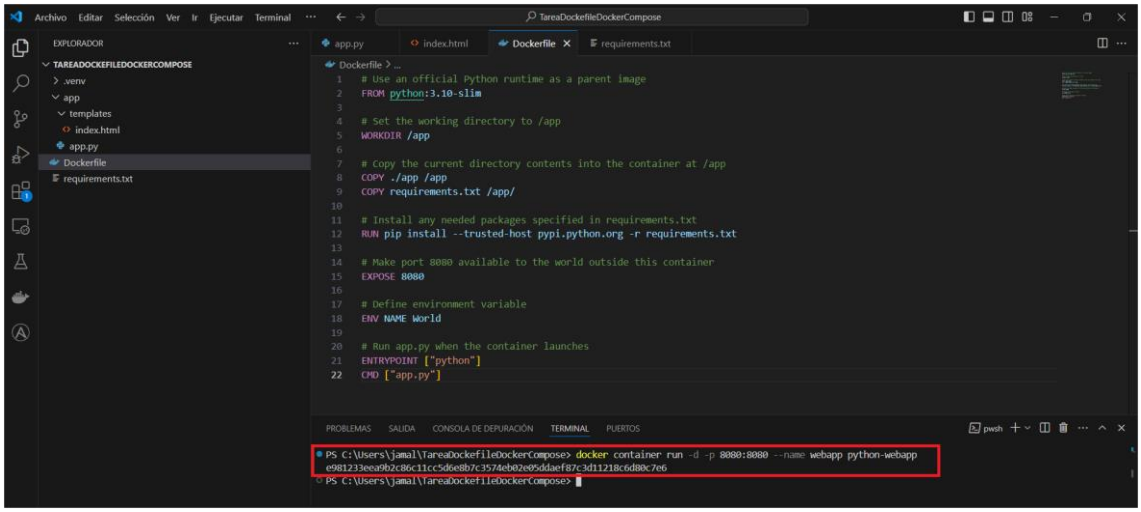
Fichero Dockerfile



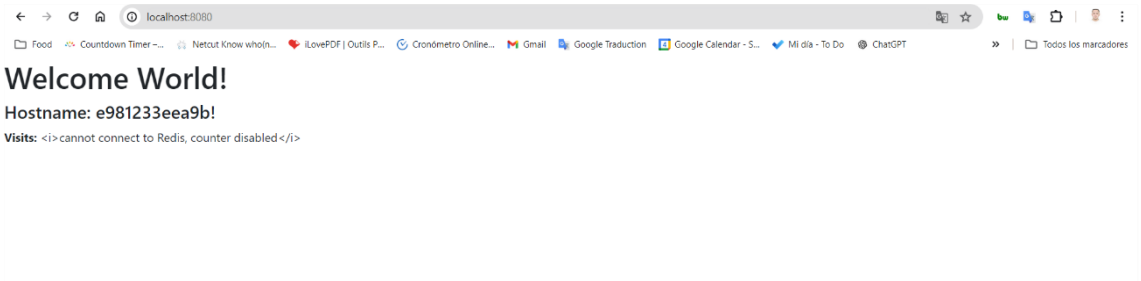
Construir la imagen



Ejecuta en un contenedor

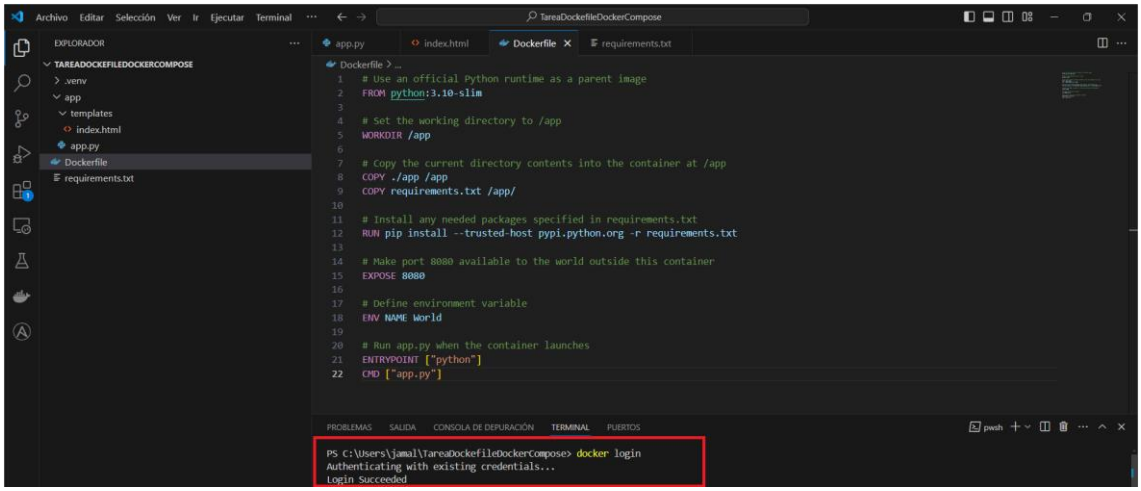


Obviamente, aún no tenemos el contenedor con el almacenamiento Redis; por lo tanto, aparece el mensaje de error.

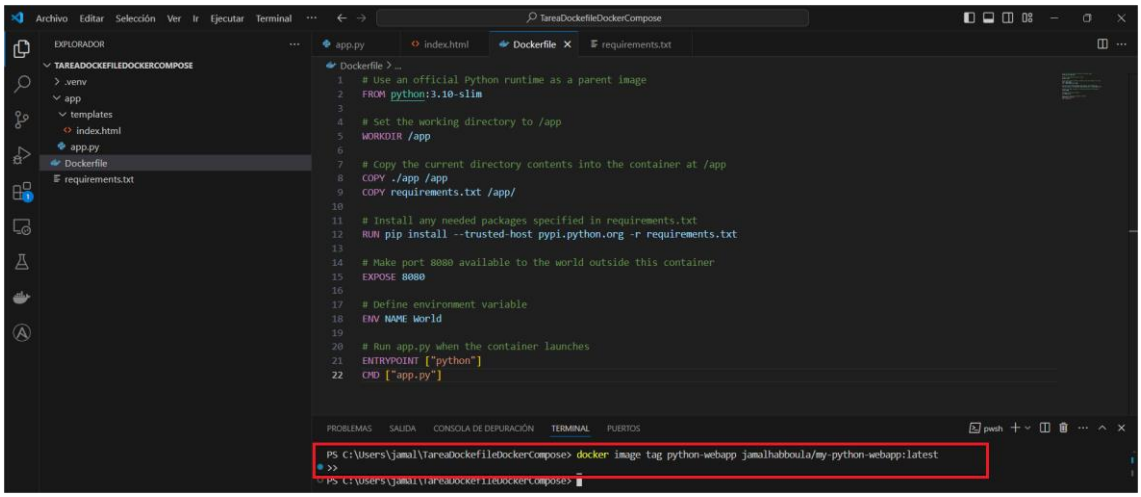


Subir la imagen a docker Hub

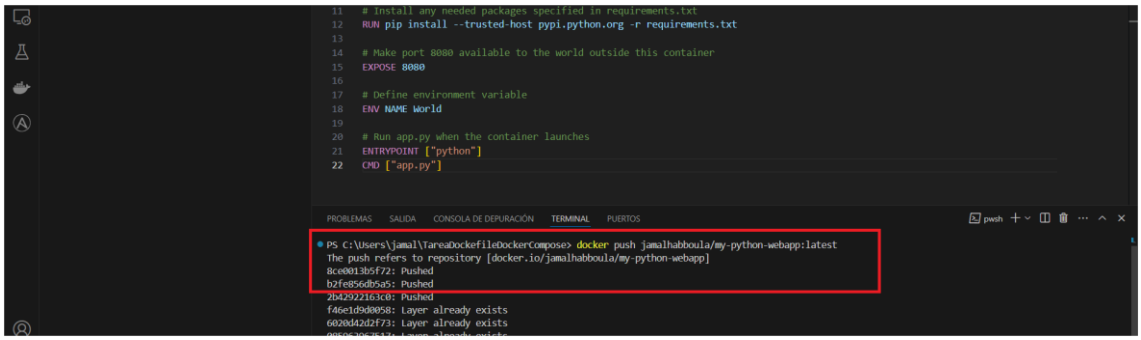
Autoriza a Docker a trabajar con tu Docker Hub, introduciendo las credenciales con:

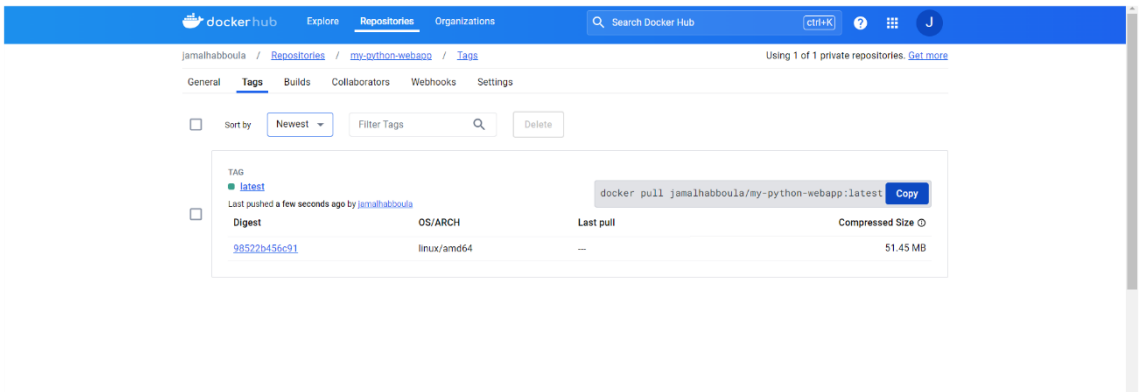


Etiqueta la imagen con tu usuario, el nombre que desees darle a la imagen y un tag (será lastest si no incluyes ninguno).

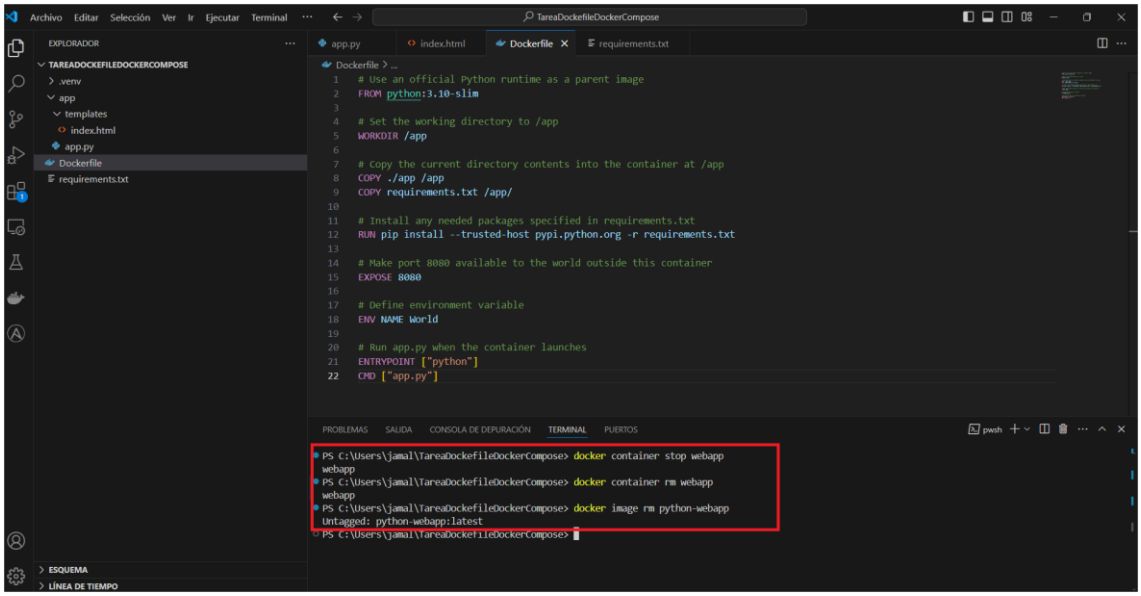


Sube la imagen al DockerHub





Elimina el contenedor y la imagen del host local

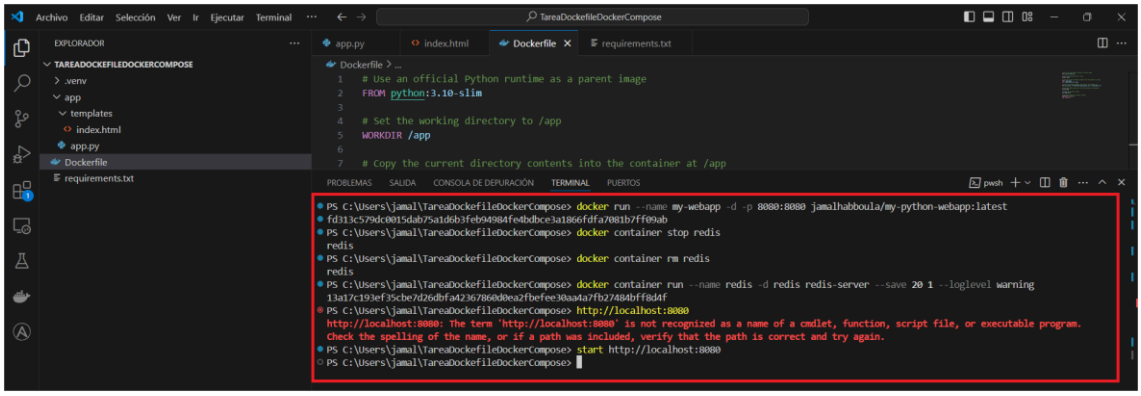


Iniciar un contenedor con el almacenamiento Redis

Si lo necesitas, Busca en el Docker Hub para más información sobre la imagen Redis.

```
PS C:\Users\jamal\TareaDockefileDockerCompose> docker container run --name redis -d redis redis-server --save 20 1 --loglevel warning
9484d3455f77ed7bc5d36859f86a714a6973c5a106289bf2ece5918c1c791da1
PS C:\Users\jamal\TareaDockefileDockerCompose>
```

Iniciar contenedor de la aplicación



Creamos el volumen

```
# Use an official Python runtime as a parent image
FROM python:3.10-slim

# Set the working directory to /app
WORKDIR /app

# Copy the current directory contents into the container at /app
```

```
PS C:\Users\jamal\TareaDockefileDockerCompose> docker volume create redis-vol
redis-vol
PS C:\Users\jamal\TareaDockefileDockerCompose> docker volume inspect redis-vol
[
    {
        "CreatedAt": "2024-06-06T12:00:42Z",
        "Driver": "local",
        "Labels": null,
        "Mountpoint": "/var/lib/docker/volumes/redis-vol/_data",
        "Name": "redis-vol",
        "Options": null,
        "Scope": "local"
    }
]
```

```
See 'docker run --help'.
PS C:\Users\jamal\TareaDockefileDockerCompose> docker container stop redis
redis
PS C:\Users\jamal\TareaDockefileDockerCompose> docker container rm redis
redis
PS C:\Users\jamal\TareaDockefileDockerCompose> docker run --name redis -d -v redis-vol:/data redis redis-server --save 20 1 --loglevel warning
b3b66af80478126c19b5a3e5ea98fd3ec146405c6cb5739dc55b764244eae10d
PS C:\Users\jamal\TareaDockefileDockerCompose>
```

Docker compose

```
PS C:\Users\jamal\TareaDockefileDockerCompose> docker compose version
Docker Compose version v2.24.6-desktop.1
```

Crear el fichero docker-compose.yml

```yml
version: '3.8'
services:
  redis:
    image: redis
    container_name: redis
    volumes:
      - redis-vol:/data

  web-app:
    image: jamalhabboula/my-python-webapp:latest
    container_name: python-app
    ports:
      - "8081:8080"
    depends_on:
      - redis

volumes:
  redis-vol:
    external: true
```

```
PS C:\Users\jamal\TareaDockefileDockerCompose> docker-compose up -d
>>
 [+] Running 2/2
  ✓ Container redis       Running                                          0.0s
  ✓ Container python-app  Started                                          2.4s
PS C:\Users\jamal\TareaDockefileDockerCompose> docker ps
CONTAINER ID   IMAGE                          COMMAND                  CREATED          STATUS          PORTS                    NAMES
2ed40b3fdfa1   jamalhabboula/my-python-webapp:latest   "python app.py"          13 seconds ago   Up 10 seconds   0.0.0.0:8081->8080/tcp   python-app
3f304f5d53e5   redis                          "docker-entrypoint.s…"   About a minute ago   Up About a minute   6379/tcp                 redis
fd313c579dc0   jamalhabboula/my-python-webapp:latest   "python app.py"          37 minutes ago   Up 37 minutes   0.0.0.0:8080->8080/tcp   my-webapp
PS C:\Users\jamal\TareaDockefileDockerCompose> docker-compose down
 [+] Running 3/3
  ✓ Container python-app                  Removed                          10.4s
  ✓ Container redis                       Removed                           0.6s
  ✓ Network tareadockefiledockercompose_default  Removed                   0.3s
PS C:\Users\jamal\TareaDockefileDockerCompose>
```